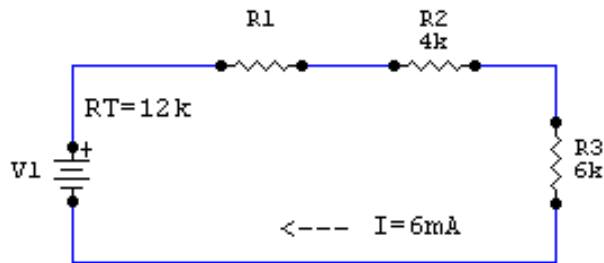
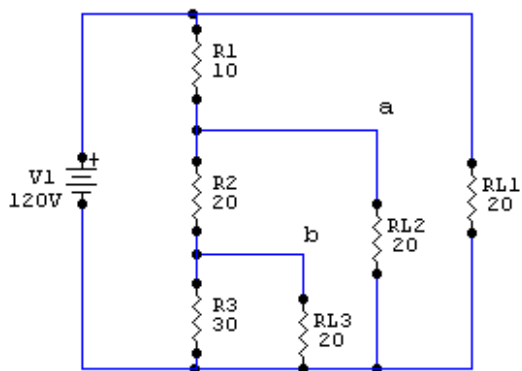


1. Ohm's Law:



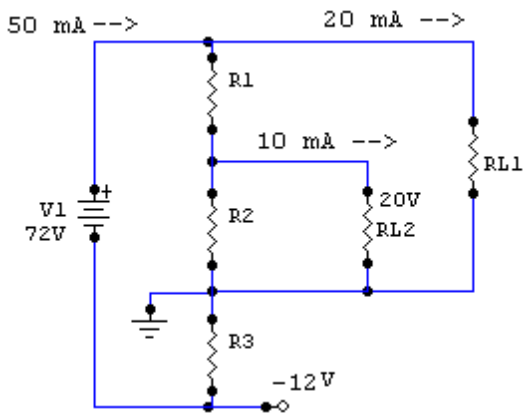
For the circuit shown, on the left, determine the values of the voltage $V1$ and the resistor $R1$

2. Voltage Divider Rule:



For the circuit shown on the left, determine the values of the voltages at points 'a' and 'b'. Determine also the current drawn from the battery.

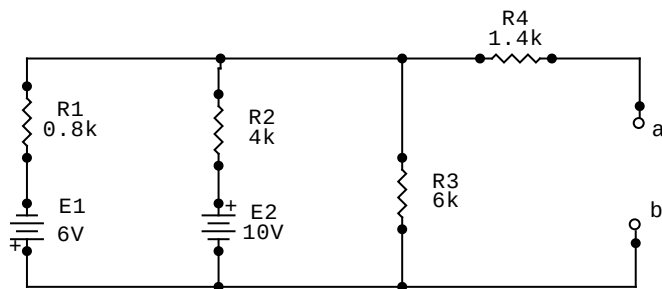
3. Kirchhoff's Current Law:



For the circuit shown on the left, determine the values of the resistors $R1$, $R2$ and $R3$.

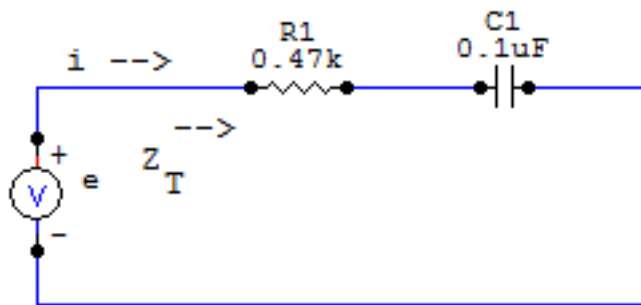
Can 2 W resistors be used in the design?

4. Thévenin, Superposition:



For the circuit shown, find the Thévenin equivalent circuit between points 'a', 'b'.

5. AC Circuits:



$e = 20 \sin(\omega t)$, and $f = 1 \text{ kHz}$
[Note that $\omega = 2\pi f \text{ rad/sec}$]

- Determine Z_T
- Find the circuit current, i .
- Calculate V_R and V_C
- Draw the phasor diagram for e , V_R , V_C

6. The R-C Circuit:

An R-C circuit has $R = 10 \text{ k}\Omega$ and $C = 0.1 \text{ }\mu\text{F}$, and $V_b = 5 \text{ V}$.

- Calculate the time constant.
- Calculate the voltage across the capacitor at $t = 0.5\tau$.
- After the capacitor is fully charged, the input side of the circuit is disconnected from the battery and connected to Ground (switch position 2 in the diagram in the Week 1 notes). Calculate the capacitor current after 0.2 ms .

7. Complex Numbers:

Determine the value of the expression in the form $A + jB$:

$$\frac{(6 + j5)(10 - j6)}{(4 - j3)(5 + j5)}$$